

Poisson Regression

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- What is Poisson Regression?
- Why do we need it?
- Poisson Theory (PMF, CDF)
- Baby Example: Crabs
- Baby Example: Insect Sprays
- Real-World Application: Weather
- Poisson for Rates and Exposure
- Summary and Key Takeaways

Poisson regression is a type of generalized linear model used to model count data, where the response variable follows a Poisson distribution and represents the number of times an event occurs in a fixed interval of time or space.

Examples:

- Hourly customer arrivals at a store
- Typos on a page
- Defects in a batch of products

Traditional linear regression fails at modeling count data for the following reasons:

- Count outcomes are non-negative integers, which are often right-skewed and violate assumptions of normality
- Linear models' predicted values can be negative, which doesn't make sense for count data
- Linear regression does not account for overdispersion (variance exceeds mean)

Poisson regression addresses these limitations by assuming that: $Y_i \sim \text{Poisson}(\lambda_i)$, where $\lambda_i > 0$



Poisson regression is suitable for count data as well as rate data, where a rate refers to the number of events per unit of time, space, or population.



- Count outcomes are non-negative integers
- Mean is equal to variance
- Events are independent
- The log of the mean rate must be a linear function of x

Assumption	Description
Non-negative integers	The response variable must be count data (0, 1, 2, ...)
Mean = Variance (Equidispersion)	Variability in the data should match the average count
Independence	Each observation/event is assumed to occur independently of others
Log link function	The log of the expected count must be linear in the predictors

Table 1: Key assumptions of Poisson regression

- Assumes equidispersion, i.e., mean is equal to variance, which is not common in real-world data
- Does not perform well with excess zero data points



The Poisson distribution is a discrete probability distribution that expresses the likelihood of a given number of events occurring in a fixed interval of time, space, or observation window, **assuming a known constant mean rate** λ and independence between events.

It is defined by the **Probability Mass Function (PMF)**:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

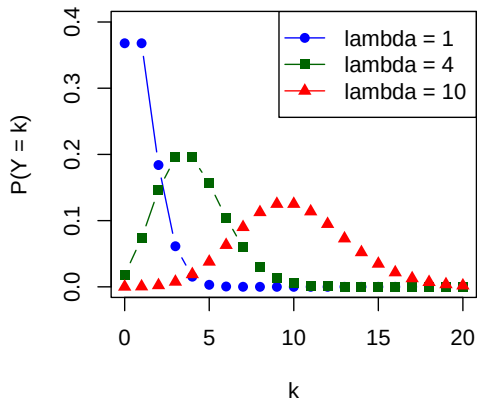
- Here, λ is both the **mean** and **variance** of the distribution:

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda$$

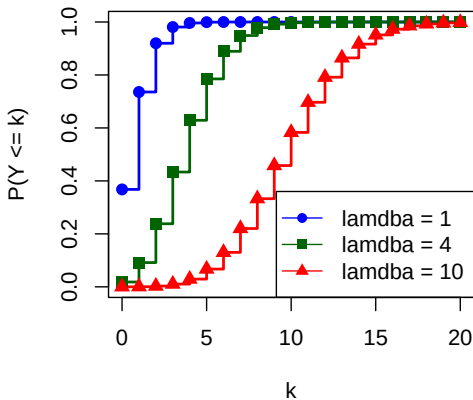
The **Cumulative Distribution Function (CDF)** gives the probability that the count is less than or equal to a particular value k :

$$P(X \leq k) = e^{-\lambda} \sum_{j=0}^k \frac{\lambda^j}{j!}$$

Poisson PMF



Poisson CDF





PMF: Probability Mass Function

- The **PMF** shows the probability of observing exactly k events.
- Smaller λ values concentrate mass near 0 (e.g., rare events).
- Larger λ values spread the distribution and approach a bell shape.

CDF: Cumulative Distribution Function

- The **CDF** shows the cumulative probability $P(Y \leq k)$.
- It rises faster as λ increases.
- Helps visualize the **likelihood of not exceeding a certain number of events**.

These plots give intuition into how count data behaves before applying any model — and help us understand **why a log-linear approach is needed** in Poisson regression.



Poisson regression extends the Poisson distribution by using predictors to estimate $\log(\lambda_i)$:

$$\log(\lambda_i) = \beta_0 + \beta_1 x_i$$

This implies:

$$\lambda_i = \exp(\beta_0 + \beta_1 x_i)$$



Let us compute the predicted values of λ_i at x_1 and x_2 for a single-regressor model:

$$\log(\lambda_1) = \beta_0 + \beta_1 x_1$$

$$\log(\lambda_2) = \beta_0 + \beta_1 x_2$$

By subtracting our first term from the second, we obtain:

$$\log(\lambda_2) - \log(\lambda_1) = \beta_1(x_2 - x_1)$$

Now, consider two values of x such that $x_2 = x_1 + 1$. When we plug this in, we find that:

$$\log(\lambda_2) - \log(\lambda_1) = \beta_1$$

Thus, the model's coefficient can be interpreted as the increase in the logarithm of the count of the outcome variable given an increase of 1 in the independent variable. In other words, a change in x has a multiplicative effect on λ .

The crabs dataset in the glm2 package contains observations on horseshoe crabs' physical traits and mating patterns.

Satellites	Width	Dark	GoodSpine	Rep1	Rep2
8	28.3	no	no	2	2
0	22.5	yes	no	4	5
9	26.0	no	yes	5	6
0	24.8	yes	no	6	6
4	26.0	yes	no	6	8
0	23.8	no	no	8	8

Let us use this dataset to build a simple Poisson regression model with Width as the independent variable and Satellites as the response variable to answer, how does the width of a female crab affect her expected number of satellites?

We can achieve this using the `glm()` function with `Satellites` as a function of `Width` and family set to `"Poisson"`:

Call:

```
glm(formula = Satellites ~ Width, family = "poisson", data = crabs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.30476	0.54224	-6.095	1.1e-09 ***
Width	0.16405	0.01997	8.216	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 632.79 on 172 degrees of freedom
Residual deviance: 567.88 on 171 degrees of freedom
AIC: 927.18

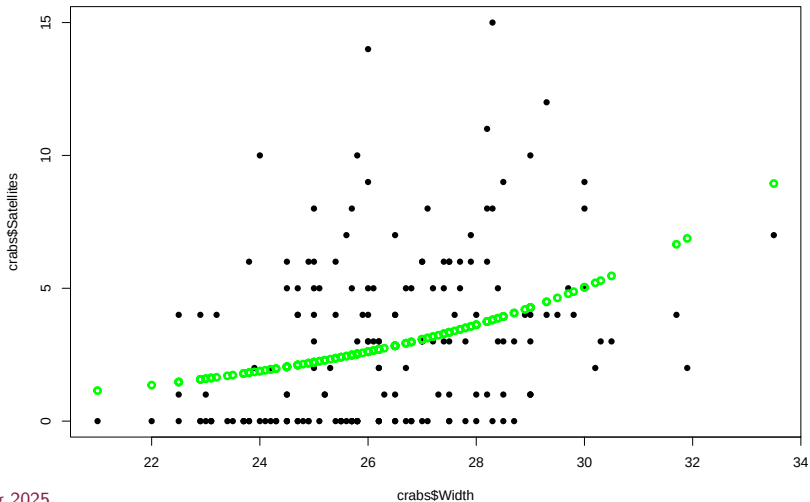
Number of Fisher Scoring iterations: 6



Thus, we have our model:

$$\lambda_i = \exp(-3.30476 + 0.16405x_i)$$

Note that $E[X]$ does not have to be an integer.





The *InsectSprays* data set contains the number of insects in agricultural plots treated with different types of insecticides. This is a classic example of modeling environmental or agricultural data using Poisson regression.

We will model the number of insects killed using Poisson regression. Our goal: **Estimate and compare the effectiveness of different spray types** in reducing insect populations.

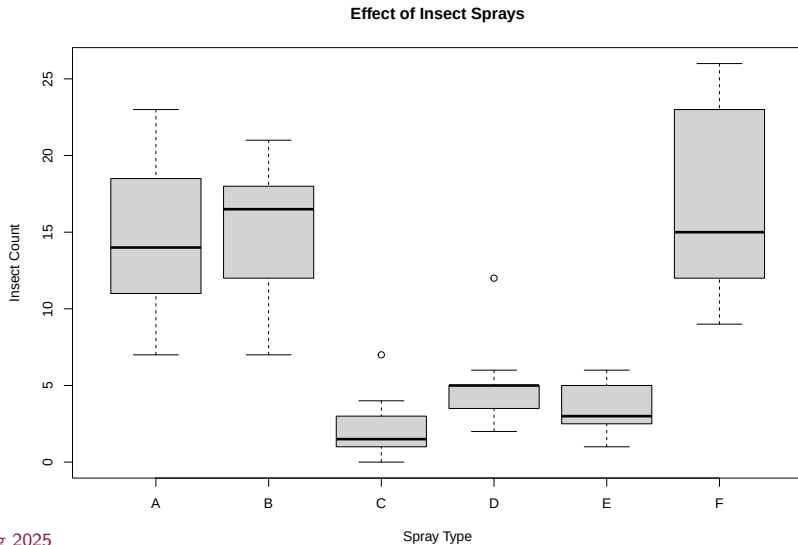
```
head(InsectSprays) %>% kable()
```

count	spray
10	A
7	A
20	A
14	A
14	A
12	A

- ***count***: Number of insects in a plot (response variable).
- ***spray***: Factor variable indicating the spray type (A-F).

```
tail(InsectSprays) %>% kable()
```

	count	spray
67	13	F
68	10	F
69	26	F
70	26	F
71	24	F
72	13	F





Given the boxplot in the previous slide:

- Spray C, D, and E: Clearly show they are the most effective (lowest insect count across all other observations).
- Spray A and B: Similar distributions where B has a higher **median** but also more consistency.
- Spray F: Has the **highest variability** and **highest maximum counts** but also the highest **median**.

```
aggregate(count ~ spray, data = InsectSprays, FUN = mean) %>% kable()
```

spray	count
A	14.500000
B	15.333333
C	2.083333
D	4.916667
E	3.500000
F	16.666667

Call:

```
glm(formula = count ~ spray, family = poisson, data = InsectSprays)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.67415	0.07581	35.274	< 2e-16 ***
sprayB	0.05588	0.10574	0.528	0.597
sprayC	-1.94018	0.21389	-9.071	< 2e-16 ***
sprayD	-1.08152	0.15065	-7.179	7.03e-13 ***
sprayE	-1.42139	0.17192	-8.268	< 2e-16 ***
sprayF	0.13926	0.10367	1.343	0.179

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 409.041 on 71 degrees of freedom
Residual deviance: 98.329 on 66 degrees of freedom
AIC: 376.59

Number of Fisher Scoring iterations: 5



We have these two functions in our arsenal — what are they, and how can we use them?

dpois(k, λ): Gives the *probability* of observing exactly k events when the expected count is λ

ppois(k, λ): Provides the *cumulative probability* $P(X \leq k)$



We will examine **Spray A** (our baseline group) for our functions. We estimate the **expected number of insects** to be λ_A



- 1 Calculate mean count (which we did) for Spray A

```
lambda_A <- mean(InsectSprays$count[InsectSprays$spray == "A"]) # we got 14.50
```

- 2 Ask the question: What is the probability of getting exactly 10 insects in a plot?

```
dpois(10, lambda = lambda_A) # Looking for  $P(X = 10)$ 
```

```
[1] 0.05710143
```

- 3 Question: What is the chance we happen to see 10 or fewer insects in our data?

```
ppois(10, lambda = lambda_A) # Looking for  $P(X \leq 10)$ 
```

```
[1] 0.1448608
```

These help us understand:

- How likely a certain outcome is, **given our Poisson model**
- Why Poisson regression is ideal for count-based outcomes



Poisson regression is also suitable for rate data, in which a rate refers to a count of events divided by a fixed interval of space or time, known as *exposure*. In Poisson regression, this is handled with an *offset*.

Let us define a given rate as:

$$\text{Rate}_i = \frac{\lambda_i}{\text{exposure}_i}$$

To model rates, we rewrite the log-linear model as:

$$\log\left(\frac{\lambda_i}{\text{exposure}_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ik}$$

This is equivalent to:

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \log(\text{exposure}_i)$$

Where the added term, $\log(\text{exposure}_i)$, refers to the model's **offset**.



```
# Simulate realistic data
set.seed(123)
n <- 100
region <- paste0("Region_", 1:n)
population <- round(runif(n, 1000, 10000)) # exposure variable
pollution <- runif(n, 0, 1) # continuous predictor
cases <- rpois(n, lambda = exp(1 + 0.8 * pollution + log(population))) # count outcome

# Create data frame
df <- data.frame(region, population, pollution, cases)
head(df) %>% kable()
```

region	population	pollution	cases
Region_1	3588	0.5999890	15672
Region_2	8095	0.3328235	28723
Region_3	4681	0.4886130	18971
Region_4	8947	0.9544738	52062
Region_5	9464	0.4829024	37848
Region_6	1410	0.8903502	7853

```
model_naive <- glm(cases ~ pollution, family = poisson, data = df)
summary(model_naive)
```

Call:

```
glm(formula = cases ~ pollution, family = poisson, data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	9.683565	0.001554	6230.8	<2e-16 ***
pollution	0.654858	0.002512	260.6	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 623449 on 99 degrees of freedom
Residual deviance: 555516 on 98 degrees of freedom
AIC: 556692

Number of Fisher Scoring iterations: 4



```
model_offset <- glm(cases ~ pollution + offset(log(population)), family = poisson, data = df)
summary(model_offset)
```

Call:

```
glm(formula = cases ~ pollution + offset(log(population)), family = poisson,
     data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.999843	0.001544	647.4	<2e-16 ***
pollution	0.799924	0.002493	320.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

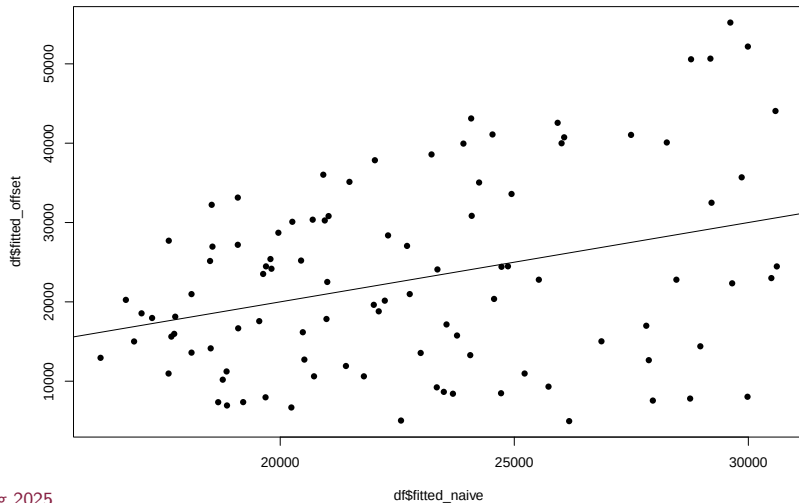
Null deviance: 102755.97 on 99 degrees of freedom
Residual deviance: 92.26 on 98 degrees of freedom
AIC: 1268.4

Number of Fisher Scoring iterations: 3

Here, we compare predictions. The offset model corrects for population differences.

region	fitted_naive	fitted_offset
Region_1	23776.97	15758.558
Region_2	19960.63	28712.234
Region_3	22104.52	18806.606
Region_4	29989.71	52178.357
Region_5	22022.01	37849.716
Region_6	28756.46	7811.877

- The naive model ignores exposure, treating all regions as if they have the same population size, producing biased estimates
- By using population as an offset and adjusting for exposure, we can more accurately estimate the rate of cases per person



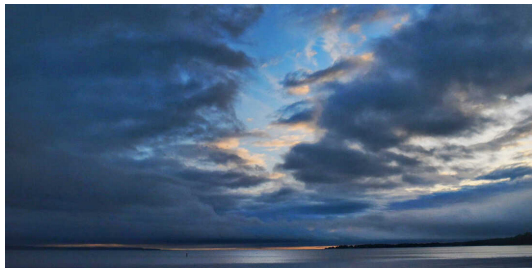


- The scatterplot shows that the two models give different results. The offset model adjusts for exposure, so its predictions often shift below the naive ones.
- Population is the exposure. Larger populations naturally have more cases.
- The naive model fails to account for exposure differences. Using an offset in the model adjusts for this and models case rates, leading to more accurate interpretation of the effect of pollution on disease occurrence.

We used a data set that contained daily weather data from a diverse set of capital cities throughout the world to predict how many “cloudy days” are in a week’s worth of data.

Useful for:

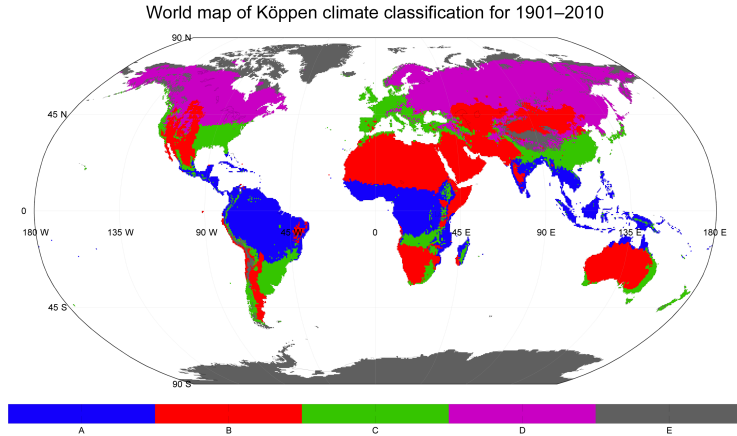
- Solar Energy
- Tourism
- Film, TV, Photography





- 60,000 daily weather records
- Data from 248 Capital cities from around the world
- From August 2023 to today
- Key Features:
 - `temperature_fahrenheit`: Air temperature (°F) — affects saturation and cloud formation
 - `humidity`: Relative humidity (%) — higher humidity increases the likelihood of cloud formation
 - `pressure_in`: Atmospheric pressure (inHg) — lower pressure systems often support rising air and cloud development
 - `wind_mph`: Wind speed (mph) — included as an alternate unit for interpretability

We needed a way to take climate into account without using the exact location to prevent over fitting, so we matched each location to their Köppen Climate Classification and joined it to the dataset to encode long term climate patterns.





We went through every week in the dataset and counted how many cloudy days (coverage == 100) were in that week and averaged important features for predicting cloudy days.

```
weekly_data <- df2 %>%  
  group_by(year, week,  
            kg_group, kg_subgroup1, kg_subgroup2) %>%  
  summarise(  
    CloudyCount    = sum(cloud == 100, na.rm = TRUE),  
    avg_temp_f     = mean(temperature_fahrenheit, na.rm = TRUE),  
    avg_humidity    = mean(humidity, na.rm = TRUE),  
    avg_pressure    = mean(pressure_in, na.rm = TRUE),  
    avg_wind_kph    = mean(wind_kph, na.rm = TRUE),  
    .groups         = "drop"  
  )
```



We used the **GLM function** to run Poisson Regression on the data.

```
model_weekly_pois <- glm(  
  CloudyCount ~  
    kg_group +  
    kg_subgroup1 +  
    ...,  
  data = weekly_data,  
  family = poisson(link = "log")  
)
```

Null deviance: 2227.7 on 875 degrees of freedom

Residual deviance: 1210.6 on 859 degrees of freedom

45.7% of the variation in cloudy-day counts is explained by our model!

Our **dispersion ratio was 1.23** which means the variance was slightly larger than the mean which is not ideal for Poisson, but still statistically valid.

$$\lambda_i = \exp(-12.584576 + -0.929033x_i + \dots)$$

■ Top Coefficients:

- avg_humidity: +3.5% cloudy days per +1% RH
- avg_wind_kph: +13.6% per +1 km/h
- avg_gust_mph: -13% per +1 mph
- kg_subgroup1s (dry-summer): ~60% fewer cloudy days
- kg_subgroup1m (monsoon): ~66% more cloudy days

- Poisson regression is designed for count data
- Use it when data are integers ≥ 0 and variance = mean
- We saw 4 examples:
 - Biological (crabs & mates)
 - Agricultural (insecticide effects)
 - Epidemiology (disease spread)
 - Environmental (cloudy days across climates)
- Offsets allow rate-based modeling
- Tools like `dpois()` and `ppois()` tie theory to real-world data

Fin